Assignment 5: Database Creation and Table Implementation

COLLEGE MANAGEMENT SYSTEM

DBMS Assignment 5

Database: CollegeManagementSystem

SQL Platform: MySQL 8.0.16 or later

Project scope: departments, faculty, students, courses, and enrollments.

Note: Query outputs described in this report are expected results from the included fictional sample data. Run the SQL script in MySQL Workbench and capture actual screenshots if your instructor requires them.

Objective

Create the CollegeManagementSystem database and implement tables using SQL DDL.

Tables and relationships

Departments stores department details. Students and Faculty each reference Departments. Courses references both Departments and Faculty. Enrollments references Students and Courses.

DDL implementation

```
CREATE DATABASE IF NOT EXISTS CollegeManagementSystem;
USE CollegeManagementSystem;

DROP TABLE IF EXISTS Enrollments;
DROP TABLE IF EXISTS Courses;
DROP TABLE IF EXISTS Students;
DROP TABLE IF EXISTS Faculty;
DROP TABLE IF EXISTS Departments;

CREATE TABLE Departments (
 department_id INT AUTO_INCREMENT PRIMARY KEY,
 department_name VARCHAR(100) NOT NULL UNIQUE,
 office_location VARCHAR(100) NOT NULL
);
CREATE TABLE Faculty (
 faculty_id INT AUTO_INCREMENT PRIMARY KEY,
 faculty_name VARCHAR(100) NOT NULL,
 email VARCHAR(150) NOT NULL UNIQUE,
 department_id INT NOT NULL,
 salary DECIMAL(10,2) NOT NULL CHECK (salary >= 0),
 employment_status VARCHAR(20) NOT NULL DEFAULT 'Active',
 CHECK (employment_status IN ('Active','On Leave','Retired')),
 FOREIGN KEY (department_id) REFERENCES Departments(department_id)
);
CREATE TABLE Students (
 student_id INT AUTO_INCREMENT PRIMARY KEY,
 register_no VARCHAR(20) NOT NULL UNIQUE,
 student_name VARCHAR(100) NOT NULL,
 email VARCHAR(150) NOT NULL UNIQUE,
 date_of_birth DATE,
 admission_year YEAR NOT NULL,
 department_id INT NOT NULL,
 FOREIGN KEY (department_id) REFERENCES Departments(department_id)
);
CREATE TABLE Courses (
 course_id INT AUTO_INCREMENT PRIMARY KEY,
 course_code VARCHAR(20) NOT NULL UNIQUE,
 course_name VARCHAR(120) NOT NULL,
 credits TINYINT NOT NULL CHECK (credits BETWEEN 1 AND 6),
 department_id INT NOT NULL,
 faculty_id INT NOT NULL,
 FOREIGN KEY (department_id) REFERENCES Departments(department_id),
 FOREIGN KEY (faculty_id) REFERENCES Faculty(faculty_id)
);
CREATE TABLE Enrollments (
 enrollment_id INT AUTO_INCREMENT PRIMARY KEY,
 student_id INT NOT NULL,
 course_id INT NOT NULL,
 semester TINYINT NOT NULL CHECK (semester BETWEEN 1 AND 8),
 academic_year VARCHAR(9) NOT NULL,
 grade VARCHAR(2) NOT NULL DEFAULT 'NA',
 enrollment_date DATE NOT NULL,
 UNIQUE (student_id, course_id, semester, academic_year),
 CHECK (grade IN ('O','A+','A','B+','B','C','F','NA')),
 FOREIGN KEY (student_id) REFERENCES Students(student_id),
 FOREIGN KEY (course_id) REFERENCES Courses(course_id)
);
```

Execution notes

Run the SQL script in MySQL Workbench. DROP TABLE statements allow a clean practice rebuild but remove existing data; remove these lines if you need to preserve data.

End of Assignment Report